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Citation for the original published paper (version of record):

Nilsson, A., Iqbal, S. (2026)
A Study of User-Focused Design Interventions to Counter Misinformation on Social Media
Emerging Media, 4(3): 546-568
https://doi.org/10.1177/27523543261474890

Access to the published version may require subscription.

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Original Research Article

A Study of User-Focused Design Interventions to Counter Misinformation on Social Media

Emerging Media
2026, Vol. 4(3) 546–568
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DOI: 10.1177/27523543261474890
journals.sagepub.com/home/emm
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Abstract

This study examines how social media users interpret and respond to two transparent interface nudges designed to encourage more careful engagement with news content. Drawing on dual-process theory, social norm theory, and nudge theory, the study developed an interactive Facebook-style mobile prototype with an accuracy prompt triggered during sharing and a persistent social-norm message above posts. Using an exploratory, qualitatively driven Design Science Research approach, data were collected from 24 participants through click-stream logs, post-simulation survey responses, and seven semistructured interviews with intervention-group participants. The findings show that participants often described themselves as capable of identifying misinformation while relying in practice on emotional salience, familiarity, social cues, and low-effort forms of engagement. The social-norm message was more consistently noticed, whereas the share-triggered accuracy prompt had limited reach because sharing was infrequent. Participants generally perceived the nudges as nonintrusive, and some reported moments of hesitation or reconsideration. However, the study does not provide evidence of improved accuracy discernment, belief change, or direct System 2 processing. Its contribution lies in showing how established drivers of misinformation engagement shape the reception and limits of interface nudges in a simulated mobile-feed setting. The study offers design implications for developing transparent, autonomy-preserving interventions that better align with passive and socially cued forms of platform engagement.

Keywords

fabricated news, misinformation, UI design, perceptual bias, social norms, digital behavior

Received: April 3, 2026; revised: July 21, 2026; accepted: July 21, 2026

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Introduction

Social media has fundamentally transformed how information is created, shared, and consumed by enabling rapid dissemination and broad accessibility. At the same time, these platforms have introduced significant challenges related to the spread of misinformation. Facebook, X, TikTok, and similar platforms have become an essential news source for many users (We Are Social & Meltwater, 2025). This shift has weakened the role of traditional journalistic gatekeepers by moving quality control from trained journalists to ordinary users, who often encounter news through platform-curated feeds and may have limited opportunity or motivation to verify content before engaging with it (Kim & Dennis, 2019; Kim et al., 2019). Drawing on cognitive psychology and behavioral design, this study examines whether small, purposeful interface interventions, also known as nudges, can disrupt automatic engagement with misinformation and prompt more reflective evaluation of online information. Misinformation disseminated through social platforms is an increasing concern, affecting public trust, decision-making, and democratic processes (Pennycook et al., 2020a, 2020b, 2021). Its effects extend to political polarization, public health behavior, and trust in science and journalism. During the 2016 U.S. presidential election, false news stories sometimes outperformed factual reporting in reach and engagement, raising concerns about their influence on public opinion (Lazer et al., 2018; Spohr, 2017).

Misinformation extends beyond politics, affecting public understanding of science, health, and climate issues. The "implied truth effect" describes how unchallenged misinformation can seem credible (Pennycook et al., 2020a). French et al. (2025) found that almost two-thirds of social media users do not fact-check the information they encounter, allowing misinformation to circulate with limited interruption. Both political and technological factors contribute to this rapid diffusion (Horner et al., 2021; Kim & Dennis, 2019). The algorithms of social platforms are designed to prioritize engaging content (which may be emotional or sensational) over some degree of accuracy (Horner et al., 2021). At the same time, users are often guided by fast, less analytical, and emotion-driven System 1 thinking rather than slower, more analytical System 2 thinking, especially in mobile-first, fast-paced settings (French et al., 2025). Although platforms provide fact-checking features, these are not always systematically applied and may be slow to react. According to Gimpel et al. (2021), a post can already be viralized by the time it has been flagged.

While prior work has established that accuracy prompts and social-norm cues can reduce misinformation sharing, this evidence is largely quantitative, lab-based, and focused on aggregate effects. Comparatively little research examines how ordinary users actually interpret and experience such nudges within a realistic mobile feed. This study addresses that gap by foregrounding users' own interpretations of two transparent nudges in a simulated Facebook-style environment. To do so, an interactive prototype simulating a common mobile social media feed was developed and piloted. The study is guided by the following research question: How do social media users interpret and respond to accuracy and social-norm nudges, and to what extent do these nudges prompt more reflective engagement with misinformation? Here, reflective engagement refers to self-reported indications of a possible shift from automatic, emotion-driven System 1 processing toward more effortful System 2 evaluation, while the interface design examined in this study refers specifically to the two nudges incorporated into the prototype. Using a qualitatively driven Design Science Research (DSR) approach, the study explores ethical, user-focused design strategies that encourage reflective engagement while preserving usability.

The contribution of this study is not to establish that emotion, overconfidence, familiarity, or social validation influence misinformation engagement, as these mechanisms are already well-documented in prior research. Rather, the study examines how these established mechanisms

condition users' interpretations of interface nudges and constrain their practical effectiveness within a simulated mobile social media feed. By focusing on users' experiences of an accuracy prompt and a social-norm message, the study contributes a qualitative account of why theoretically promising nudges may be noticed, accepted, ignored, externalized as advice for others, or weakened by native platform cues such as comments and reactions. The study therefore extends prior work on misinformation nudging by shifting attention from whether nudges can work in aggregate to how users interpret and encounter them during ordinary interaction flows.

The remainder of the article is organized as follows. Next section reviews related research, and Theoretical Framework section presents the theories used in this research work. Methodology section describes the methods for data collection and analysis. Artifact developed for this study is presented in Artifact Design section, Empirical Findings section presents the findings from the data. Artifact Evaluation section presents the results of the evaluation based on DSR criteria. Discussion and Further Research section discusses the findings, limitations, and directions for future research.

Related Research

This section outlines the main factors behind the spread of misinformation on social media and supports the development of design-based solutions at the interface level. The literature is organized into four key areas: emotional and cognitive user responses, psychological biases, how content is presented, and interface design interventions. Together, these themes help explain how misinformation spreads, how users engage with it, and what design approaches have been considered to counter its impact.

Emotional and Cognitive Factors in User Behaviour

Emotional states and cognitive inattention significantly influence how users engage with misinformation on social media. Dabbous and Barakat (2023) found that both positive and negative emotions increase the likelihood of sharing fake news. Positive emotions may encourage uncritical social sharing, while negative emotions, especially during crises like COVID-19, can contribute to information overload and reinforce existing beliefs. Horner et al. (2021) observed that emotionally charged content, such as anger, disgust, or happiness, can drive engagement or toxic responses, although content that clashes with users' political views may also be ignored. Pennycook et al. (2021) argue that misinformation often spreads due to inattention rather than deliberate bias. Balakrishnan et al. (2021) further support this view, indicating that misinformation is frequently shared for nonmalicious, psychologically driven reasons such as perceived helpfulness or entertainment rather than intentional deception. In prior studies, users struggled more to distinguish truth from falsehood when deciding what to share than when asked directly about content accuracy, suggesting that social media contexts can distract from critical evaluation. Dual-process theory, as used by Moravec et al. (2020), explains this behavior through two cognitive systems: fast, emotion-driven System 1 (dominant on social media) and slower, analytical System 2. Most misinformation is processed via System 1, reinforcing biases like confirmation bias, even when System 2 is engaged. Even without explicit reference to the theory, other studies highlight cognitive dissonance in avoiding factual but emotionally uncomfortable content. Kim and Dennis (2019) found that users were less likely to engage with belief-challenging news. Gimpel et al. (2021) and Andı and Akesson (2021) suggest that social interventions can trigger dissonance and promote reflection.

These findings stress the need for interface designs that can interrupt automatic engagement and encourage more reflective evaluation. The shift from intuitive to analytic processing is not automatic; it can be provoked by cognitive friction. A salient interruption, prompt, or unexpected cue momentarily raises processing effort and suspends habitual, heuristic responding, opening a brief window in which analytic (System 2) evaluation becomes possible. This is precisely why accuracy prompts are positioned at decision points such as the moment of sharing: the interruption coincides with the action it is meant to make users reconsider. Understanding this mechanism clarifies both the promise and the fragility of interface nudges, since the reflective window they create is short-lived and easily overridden by emotional salience.

Psychological Biases and Misinformation Engagement

While emotions capture attention, psychological biases shape how users evaluate and share misinformation (French et al., 2025). Inattention leads users to rely on intuition and cognitive shortcuts, often resulting in poor judgments. French et al. (2025) identified five psychological biases: confirmation bias, framing, overconfidence, anchoring, and herd behavior, that shape users' engagement with misinformation. Confirmation bias, the tendency to favor information aligning with existing beliefs, is particularly influential and reinforced by the hedonic, fast-scrolling nature of social media (Kim & Dennis, 2019). Even when users encountered verifiably true headlines that contradicted their views, they often rejected them (Moravec et al., 2019). These findings highlight how rapid, intuitive processing, and cognitive biases hinder the effectiveness of logic-based corrections such as fact-checking or warning labels.

Other relevant biases include the framing effect, which shapes how users interpret information depending on its presentation (French et al., 2025). In a Facebook post, Kim and Dennis (2019) made the source more visually prominent than the headline, thus attempting to prompt the users to view the source prior to the information in an attempt to shape beliefs among users. The framing effect is also relevant to the present study, as content was presented within a modified interface context. Overconfidence bias may lead users to overestimate their ability to distinguish credible from false content. Anchoring bias may make users rely too heavily on initial information in a news post, such as the headline or image, even when that information is later challenged or corrected. Anchoring supports confirmation bias when initial information supports beliefs. Herd behavior may increase trust in content that appears popular, a tendency reinforced by engagement-based platform algorithms. Previous research also highlights familiarity bias, where repeated exposure to misinformation increases its perceived truthfulness even after correction (Lazer et al., 2018; Rhodes, 2022). These biases are automatic and therefore, even skeptical users can internalize falsehoods without realizing that they are doing it because they see it so often.

Despite increasing awareness of misinformation, psychological biases continue to influence what users consider believable. These biases also provide a rationale for interface-based nudges. Accuracy prompts are intended to counter tendencies such as inattention and overconfidence by encouraging users to pause and evaluate content more carefully, while social-norm cues may help counter herd behavior and false consensus effects by drawing attention to alternative social expectations. Rather than eliminating bias altogether, such interventions seek to interrupt automatic cognitive shortcuts and promote more reflective evaluation of online information.

Platform Design and the Presentation of Misinformation

The design of social media platforms fundamentally shapes how misinformation is spread and perceived. Platforms such as Facebook and Twitter, now X, present all content—reputable news,

fabricated stories, and opinion pieces—in similar visual formats, creating the impression that all sources are equally credible (Rhodes, 2022). Algorithmic ranking and personalized feeds further reinforce exposure to content similar to what users already engage with. Spohr (2017) describes "filter bubbles," personalized environments that insulate users from opposing viewpoints. This leads to "echo chambers," where users repeatedly encounter like-minded perspectives, reinforcing existing beliefs and limiting exposure to corrective information (Lazer et al., 2018). Such environments strengthen confirmation bias by repeatedly validating preferred narratives. Studies have also shown that repeated exposure can increase familiarity and perceived truthfulness, creating a cycle in which false information becomes more believable simply because it is seen more often (Rhodes, 2022).

Misinformation can thrive when platform incentives prioritize engagement over accuracy, rewarding sensational or emotionally charged content (Rhodes, 2022). Kim and Dennis (2019) found that, offline, people typically evaluate the source before the message, whereas on social media, headlines often appear first, encouraging snap judgments linked to the biases discussed earlier. Their study also showed that when users were prompted to view the source first, they became more critical of content quality.

Therefore, the social media environment may both enable and amplify misinformation through design choices that prioritize visibility, repetition, and virality over credibility and context—an essential consideration for this study, as it defines the setting where design interventions must operate. Hemphill and Banerjee (2021) show that platform self-regulation remains limited, as engagement-driven incentives continue to shape design decisions despite growing regulatory pressure.

User Interface Interventions

Recent studies examine interventions that steer user behavior without restricting choice, such as social-norm messages, credibility indicators, and fact-checking prompts (Andı & Akesson, 2021; Bhuiyan et al., 2021; Moravec et al., 2020). Nudging, understood as subtle design choices that influence behavior while preserving autonomy, is one strategy in this domain (Bhuiyan et al., 2021; Caraban et al., 2019). Social-norm messages, presented as injunctive or descriptive cues, have been shown to reduce fake-news sharing and encourage reflection (Andı & Akesson, 2021; Gimpel et al., 2021). These approaches guide users toward more thoughtful engagement without labeling actions as right or wrong, though they must be applied carefully, as social-norm cues can inadvertently prompt users to report accurate content (Gimpel et al., 2021).

Bhuiyan et al. (2021) developed *NudgeCred*, which labels news sources according to credibility. These System 1 cues support rapid judgments but can lose influence through habituation. Similarly, Kim et al. (2019) found that low source ratings decreased believability more than high ratings increased it, yet confirmation bias still shaped responses. To counter such limitations, researchers propose combining Systems 1 and 2 strategies. Integrating fast, visual cues with concise explanations improves truth discernment (Moravec et al., 2020; Pennycook et al., 2021). Ecker et al. (2020) demonstrated that short refutations outperform false-label warnings and avoid backfire, while Konstantinou et al. (2025) stress the need for adaptable nudges that reflect users' mental states and sustain engagement over time. Social feedback also enhances credibility judgments: Prike et al. (2024) showed that badges and peer comments improved discernment and reduced engagement with misinformation.

However, despite positive outcomes, scalability remains a challenge. Gurgun et al. (2024) indicate that users often hesitate to challenge or reflect on misinformation due to social concerns and

perceived effort, even when they recognize potential inaccuracy, constraining the effectiveness of many intervention strategies. Butler et al. (2024) highlight the need for interventions effective beyond laboratory settings. Bhuiyan et al. (2018) note that browser-based solutions are difficult to implement on mobile platforms, and Ghesla et al. (2019) caution that overly strong nudges may produce unintended behavioral spillovers.

Overall, interface-level interventions that combine automatic and reflective cues, leverage social norms, and avoid coercion appear most promising. Their design must balance usability, minimize fatigue, and preserve user trust. Chaudhuri et al. (2024) frame misinformation as a multilevel problem involving users, platforms, and institutions, and identify user-empowerment through interface-level design as a promising yet underexplored intervention strategy.

Beyond demonstrating effects, it is important to consider how users perceive interface labels themselves. Credibility indicators and warning labels are frequently overlooked during fast scrolling, are sometimes read as directed at "other" users and can lose force through habituation (Bhuiyan et al., 2021; Kim & Dennis, 2019). A central, under-examined tension is that descriptive social-norm cues and accuracy prompts can pull in different directions: a descriptive norm signaling that sharing is common may implicitly normalize engagement, while an accuracy prompt demands individual effort against that flow. The two may therefore reinforce one another priming reflection that the prompt later activates or compete, when the normative cue makes engagement feel typical. This interactive tension, rather than the isolated efficacy of either cue, is the focus that the present study carries forward into its framework and findings.

Previous research reveals several drivers of misinformation, including emotional triggers, cognitive biases, and platform design, while identifying gaps that motivate this study. Current approaches rarely integrate cognitive theory with practical design, face scalability issues, and seldom foster sustained reflection. Moreover, relatively little attention has been paid to how users interpret interface interventions or how different nudges may reinforce or undermine one another in practice. Addressing these limitations requires interventions that align with the low-effort, entertainment-oriented mindset of social media use while encouraging brief, critical engagement. These insights inform this study's design of scalable, autonomy-preserving nudges that promote reflective engagement in social media environments.

Theoretical Framework

Rather than treating the three theories separately, this study integrates them into a single explanatory chain. Dual-Process Theory accounts for the default System 1 mode that dominates fast, mobile feeds; Social Norm Theory specifies the social context that shapes which heuristic responses feel appropriate; and Nudge Theory provides the means by which interface cues translate these cognitive and social dynamics into design that shifts users toward System 2. Crucially, the two nudges examined here are modeled as interacting rather than additive. The persistent social-norm message is expected to prime a reflective orientation that the share-triggered accuracy prompt can later activate at the decision point (a synergistic relationship). Conversely, because the social-norm message refers to responsible user behavior, it may also signal that sharing is common and thereby compete with the accuracy goal it is meant to support. Figure 1 represents these directional, synergistic and competitive links rather than a simple co-presence of constructs.

According to Dual-Process Theory, cognition in a human being is governed by two systems that work simultaneously: System 1, which is fast, intuitive, and emotion-directed, and System 2 which is slower, analytical and effortful (Moravec et al., 2020). System 1 thinking is the basis of emotional responses, for example, anger or humor. Misinformation research indicates that subtle interface

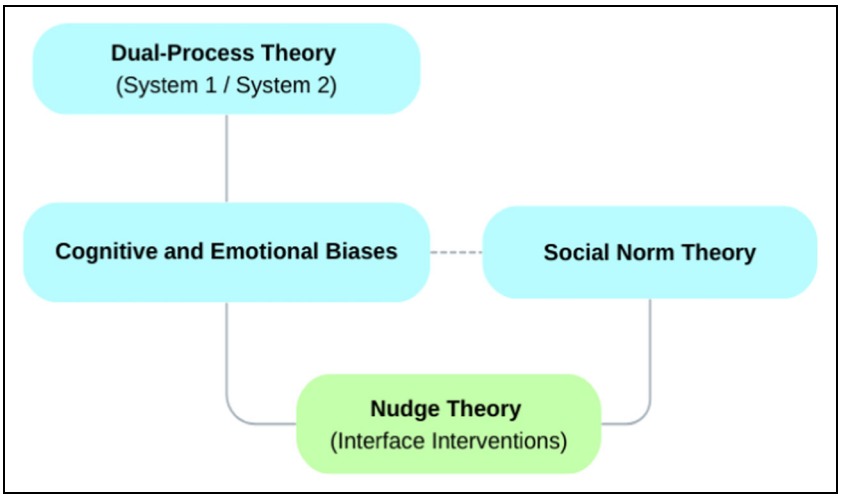


Figure 1. Theoretical framework illustrating the conceptual relationships between dual-process theory, social norm theory, and nudge theory in the context of social media interaction.

signals, such as credibility prompts or reflection reminders may facilitate System 2 involvement and enhance critical evaluation (Moravec et al., 2020; Pennycook et al., 2021). This framework highlights how interventions can interrupt automatic engagement without undermining autonomy, focusing on biases such as confirmation bias and emotional triggers that shape intuitive sharing on social media.

Social Norms Theory explains how individuals behave based on their perceptions of other users' behavior. Descriptive and injunctive norms (what the majority do and what is socially accepted) are relevant to misinformation contexts because they shape how users interpret and engage with content (Andı & Akesson, 2021; Gimpel et al., 2021). Highlighting normative behaviors, such as messages noting that most users verify news before sharing, has been shown to reduce misinformation engagement while preserving autonomy (Andı & Akesson, 2021). These cognitive and social influences jointly shape user behavior and emphasize the need for design interventions that address both mental shortcuts and social pressures.

Nudge Theory proposes structuring choice environments in ways that influence behavior without restricting available options (Bhuiyan et al., 2018). In social media settings, effectively designed interface nudges can warn users about potentially misleading information and prompt them to pause before engaging with it. This is particularly relevant in environments where attention is divided and content competes for visibility. These nudges are incorporated into interface features, which shape an automatic (System 1) or reflective (System 2) thinking. The prompts involving the consideration of accuracy activate reflective cognition, whereas visual or social ones target faster, more intuitive mechanisms. Based on the list of nudges provided by Caraban et al. (2019), in the context of interventions, there are different levels of transparency and autonomy of the user. The literature highlights that interface-level nudges must be ethically justified, transparent, and contextual, more specifically in socially sensitive domains, such as misinformation.

Together, the three theories provide a unified framework for understanding misinformation-related behavior on social media. Dual-Process Theory explains how users rely on fast, intuitive processing in information-rich environments; Social Norm Theory highlights how perceptions of

others' behavior shape engagement; and Nudge Theory provides a basis for translating these cognitive and social insights into interface-level design strategies. As illustrated in Figure 1, cognitive heuristics, social norms, and nudging strategies are conceptually interrelated and jointly inform the design of interface interventions. This theoretical framework therefore informs the practical development of nudges intended to orient users toward more reflective engagement, while preserving user autonomy.

Methodology

To avoid ambiguity, this study is explicitly framed as an exploratory, qualitatively driven DSR study. DSR provides an appropriate framework when the aim is to develop and evaluate practical design artifacts that contribute to both problem-solving and theoretical understanding (Hevner et al., 2004). While full DSR cycles typically involve multiple iterations, this study focuses on the initial artifact design and evaluation stages, establishing a foundation for future refinement. The interview data are primary; behavioral click-stream logs and survey responses are used only to contextualize and triangulate those accounts, not as the basis for controlled experimental inference. The control and intervention groups support descriptive contrast rather than hypothesis testing, and the small sample is appropriate to the exploratory, first-iteration aim of the work.

Process Overview

Following DSR principles, the study proceeded through five stages. Problem identification defined the spread of misinformation on social media, driven by cognitive bias and emotional engagement, as the central societal issue. Objective formulation set the goal of creating a user-centered artifact that could nudge engagement behavior without disrupting the user experience. Design and development involved constructing an interactive mobile prototype that simulated a social media newsfeed with embedded credibility and social-norm cues. The prototype was then demonstrated and evaluated through user interaction on the Maze platform, complemented by surveys and semistructured interviews. Finally, the communication phase involved reporting the artifact design, rationale, evaluation findings, and implications for both academic and practical audiences. While DSR commonly includes repeated design cycles, this study represents the first design–evaluation iteration and provides insights to inform future refinement.

Participants were divided into two groups: The intervention group engaged with a prototype that included nudging elements, such as credibility prompts and social-norm cues, while the control group used a version without these elements. The study examined the perceptions of the interface by the participants and their responses to the nudging elements. Although the initial aim of the study was to compare control and intervention groups, the participation was uneven, therefore limiting the direct comparison. The control group primarily provided a behavioral baseline, while the intervention group generated more qualitative data. This focus is in line with the exploratory and iterative nature of DSR where the focus may shift in response to available data and emerging insights (Hevner et al., 2004; Moravec et al., 2020).

Data Collection

Data were collected through three complementary methods: semistructured interviews, digital surveys, and behavioral click data from a simulated social media prototype. All participants first completed the simulation, followed by a postsimulation survey. A subset of intervention participants

Table 1. Interview Details by Participant ID.

Participant ID	Date	Interview Mode
Participant A	28/03/2025	In-person
Participant B	28/03/2025	In-person
Participant C	29/03/2025	Zoom
Participant D	31/03/2025	In-person
Participant E	02/04/2025	Zoom
Participant F	02/04/2025	Zoom
Participant G	03/04/2025	Microsoft Teams

was invited for interviews to explore their experiences more deeply. This data collection structure reflects a qualitatively driven design, with quantitative data used to contextualize and complement in-depth qualitative findings (Creswell & Creswell, 2018). Recruitment took place through targeted outreach on Facebook and LinkedIn and was restricted to active Facebook users to ensure contextual relevance.

The Maze platform hosted the prototype, enabling individual participation and automated logging of click-based behavioral data. The platform's single-session design reduced duplicate entries but also prevented users from restarting incomplete sessions, which contributed to minor data loss. In total, 24 participants completed the simulation (14 intervention and 10 control), 17 completed the survey, and seven from the intervention group were interviewed. Interviews lasted between 18 and 27 min and were conducted via Zoom, Teams, or in person (see Table 1). The control group was not interviewed, as the study's focus was on user experience with the nudging interventions.

Consistent with an exploratory qualitative design, sample adequacy was judged by informational saturation rather than statistical power. The seven intervention-group interviews were complemented by 17 survey responses and behavioral logs from all 24 participants, and interviewing continued until no substantially new codes emerged. The uneven group sizes resulted from recruitment yield and the platform's single-session design, which prevented restarting incomplete sessions; this is acknowledged as a limitation rather than presented as a representative sample. Interview recordings were transcribed and imported into MAXQDA, where survey and behavioral data were linked at the participant level to enable triangulation.

Data Analysis

Thematic analysis followed Braun and Clarke's (2006) six phases using a hybrid coding strategy: participant reflections were first coded inductively, after which codes were organized deductively under dual-process, social-norm, and nudge constructs, with discrepancies between the two resolved through iterative re-reading and team discussion. A reflexivity note acknowledges the researcher's active role in interpretation. Importantly, behavioral metrics (clicks, likes, unlikes, canceled shares) are treated as observable interaction patterns only; they are used to contextualize interview accounts and are not interpreted as direct evidence of internal states such as regret or hesitation. Triangulation across interviews, surveys, and behavioral logs was used to examine convergence and divergence rather than to claim quantitative generalization.

In this study, reflective engagement is defined narrowly and operationally. It does not refer to directly measured belief accuracy, truth discernment, endorsement, or System 2 cognitive processing. Instead, it refers to participants' self-reported noticing, hesitation, reconsideration, or awareness of accuracy-related cues during or after the simulation. Behavioral data, including clicks,

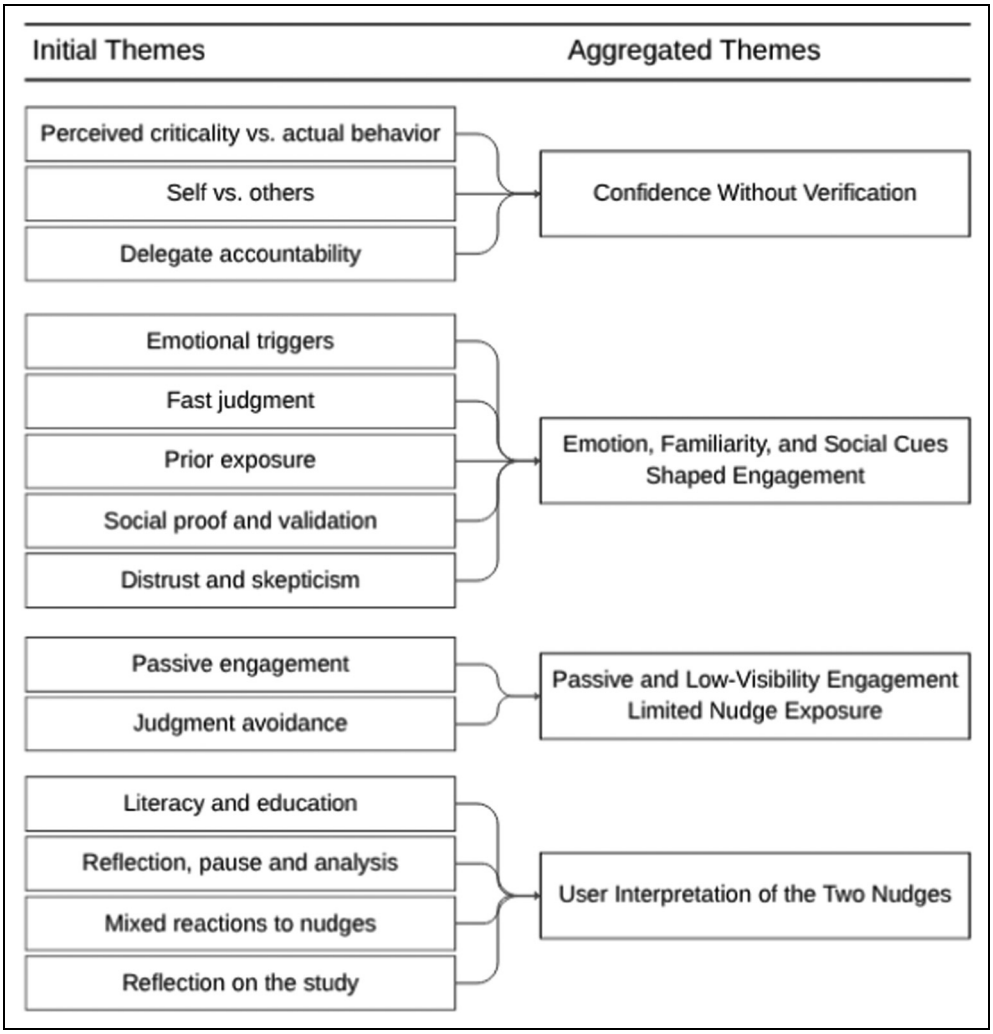


Figure 2. Thematic data structure.

canceled shares, repeated interactions, unlikes, article openings, and views of comments or reactions, were treated as observable interaction traces only. These traces were used to contextualize interview and survey accounts, not as direct evidence of internal cognitive states or accurate misinformation detection.

The study followed a primarily qualitative analytic approach supported by descriptive quantitative data. Coding was conducted in MAXQDA, and patterns were developed into thematic categories that describe users' engagement behavior, cognitive responses, and reflective engagement. Figure 2 visualizes the resulting thematic structure, illustrating how subthemes were iteratively refined and organized into aggregated themes through repeated comparison across data sources.

Survey data provided complementary insights through Likert-scale and open-ended questions. Quantitative responses were analyzed descriptively to identify general trends, such as awareness

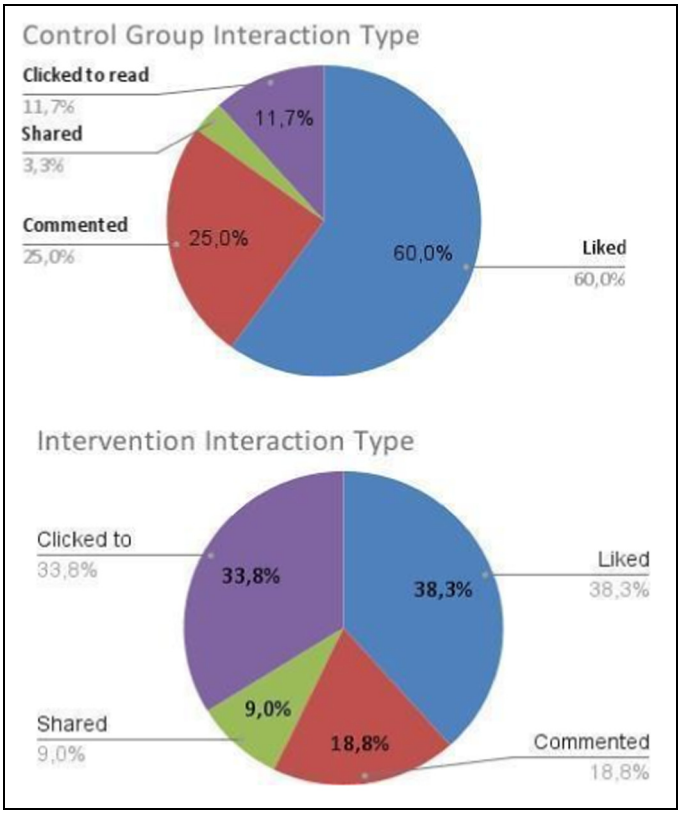


Figure 3. Distribution of interaction types by group.

of credibility, perceptions of the nudging features, and self-reported intentions regarding future behavior. These findings were integrated with the qualitative themes to enhance interpretive depth.

Behavioral click data from the Maze simulation captured engagement actions such as likes, comments, shares, and article clicks. Figure 3 presents the distribution of interaction types across the control and intervention groups, illustrating how participants engaged with posts through actions such as liking, commenting, sharing, or clicking to open articles. To summarize overall interaction levels with true and false news posts, descriptive statistics (mean, median, and mode) were calculated for both groups as reported in Table 2. One participant in the intervention group produced substantially higher interaction counts; therefore, interaction statistics for this group are reported both including and excluding this outlier to avoid distortion of central tendencies.

These data were examined to identify behavioral contrasts between the control and intervention groups, with particular attention to patterns of interaction surrounding news posts. In addition to primary interaction types, the click data were examined for additional behaviors occurring around posts. These included reversed interactions (e.g., liking and subsequently unliking a post), canceled sharing actions following the accuracy prompt, repeated interactions with the same post, and viewing comments or reactions associated with posts. Frequencies of these behaviors by group are reported in Table 3. The click data served primarily to support and contextualize the qualitative findings rather than as a basis for statistical inference.

Table 2. Descriptive Statistics of Interaction Counts With True and False News Posts by Group.

Group	Mean (False)	Median (False)	Mode (False)	Mean (True)	Median (True)	Mode (True)
Control	2.6	2	0	3.4	2	0
Intervention (with outlier)	4.3	1	0	5.1	3	0
Intervention (without outlier)	2.1	1	0	3.4	2	0

Table 3. Additional Click Behaviors Observed in Simulation.

Behavior Type	Control Group (n = 10)	Intervention Group (n = 14)
Reversed interactions (e.g., like–unlike)	8	11
Canceled share (after nudge)	–	5
Clicked to view comments	9	18
Clicked to view reactions	1	4
Multiple/repeat clicks	4	27

Together, the three data sources provided a triangulated understanding of how users interact with misinformation and respond to nudging interventions in a simulated social media environment. Behavioral metrics were used to describe observable interaction patterns within the simulation and were not interpreted as indicators of users' intentions or verification strategies.

Although Table 2 distinguishes interactions with true and false posts, these interaction counts should not be interpreted as measures of belief, endorsement, or accuracy discernment. A like, click, share attempt, or canceled share may reflect curiosity, entertainment, social interest, hesitation, or other motivations. Table 2 is therefore used descriptively to show interaction patterns across content types, not to infer whether participants correctly judged the accuracy of posts.

Artifact Design

The artifact developed for the study was an interactive prototype simulating a Facebook-style newsfeed. The feed included a curated mix of false and true headlines drawn from previous research and verified through fact-checking sources, primarily through Snopes.com due to its comprehensive, up-to-date, and frequently cited fact-checking source (e.g., Bhuiyan et al., 2021; Moravec et al., 2020). False news items reflected common misinformation themes, politics, health, and entertainment, while authentic headlines were included to increase ecological realism within the simulation. The intervention prototype incorporated two nudging mechanisms informed by prior studies:

- *Accuracy prompt*: Triggered when users attempted to share a post, this subtle message asked them to consider the accuracy of the content (see Figure 4). The design was based on Pennycook et al. (2020b, 2021), who found that reflection prompts improve discernment and reduce misinformation sharing.
- *Social norm message*: A fixed message above each post stated, "*Most responsible Facebook users think twice before sharing news"* (see Figure 5). Based on Andı and Akesson (2021), this cue appealed to socially desirable behavior while preserving autonomy.



Figure 4. Accuracy prompt.

This dual-nudge design is supported by Butler et al. (2024), who found that combining social norm messages and accuracy prompts significantly improved truth discernment. Both nudges were designed to be transparent, neutral, and nondisruptive. Following ethical frameworks from Konstantinou et al. (2025) and Caraban et al. (2019), the nudges fall within the "transparent and reflective" and "transparent and automatic" categories, engaging both intuitive (System 1) and reflective (System 2) cognition. Combining these two cognitive modes was intentional, as earlier studies indicate that pairing automatic cues with reflective prompts can enhance misinformation discernment while maintaining a positive user experience (Bhuiyan et al., 2021; Moravec et al., 2020; Pennycook et al., 2020b). The control prototype was visually identical but excluded the nudging mechanisms, allowing comparison between uninfluenced and intervention-driven engagement behaviors. Both prototypes were accessed on participants' mobile phones to better reflect typical social media use, where most interactions occur on mobile devices. The prototype was developed in Figma and implemented in Maze, a user-testing platform that recorded click-based behavioral data during the simulation. To maintain realism within practical constraints, not all interface elements were fully functional. Participants were informed that their intent to interact was the primary focus. Engagement actions such as liking, commenting, sharing, or link clicking were captured as behavioral indicators, with interaction feedback simulating authentic platform responses. Overall, the artifact balanced experimental control and ecological validity, combining theoretical grounding with practical design considerations. This dual-nudge design drew directly from cognitive and behavioral research, enabling the study of whether simple, transparent nudges can influence how users engage with misinformation without limiting their autonomy.

Empirical Findings

This section presents the empirical findings from interviews, survey responses, and click-stream logs. The purpose is not to infer belief accuracy, truth discernment, or cognitive processing directly from interaction behavior, but to describe how participants engaged with news content in the

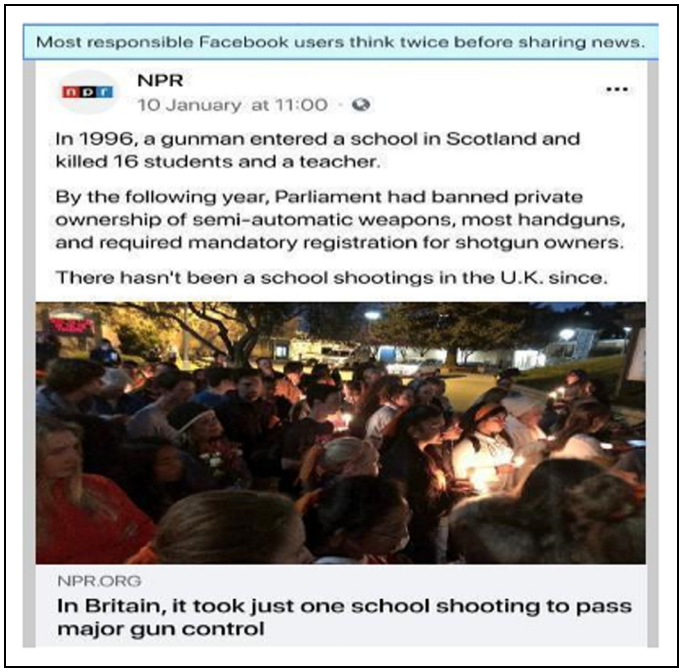


Figure 5. Social norm message.

simulated mobile feed and how they interpreted the two nudging features. The findings are organized around four themes: confidence without verification, emotion and familiarity as drivers of engagement, passive and low-visibility interaction, and user interpretation of the two nudges. Together, these themes show how established drivers of misinformation engagement shaped the visibility, interpretation, and limits of the intervention.

Confidence Without Verification

Across interviews and survey responses, participants commonly described themselves as capable of identifying misinformation and reported that they usually considered accuracy when encountering news online. However, this confidence was not consistently reflected in the interaction data. Participants rarely engaged in actions that would suggest deeper content checking, such as opening full articles or revisiting source information. Most engagement consisted of low-effort interactions, especially liking, scrolling, viewing comments, or briefly interacting with posts.

Participants also frequently framed misinformation susceptibility as a problem affecting other users rather than themselves. Older users, less digitally literate users, or users with different political views were often described as more vulnerable. Participant C stated, "I see myself among those who are more aware anyway," illustrating how some participants positioned themselves as less vulnerable than other users. This self-other distinction was also visible in responses to the nudges: several participants supported accuracy prompts and credibility cues but described them as more useful for other people than for their own behavior. This pattern suggests a gap between participants' stated confidence and their observed interaction depth within the simulation.

Emotion, Familiarity, and Social Cues Shaped Engagement

Participants' engagement with posts was strongly shaped by emotional salience, familiarity, and visible social feedback. Posts that were humorous, surprising, frightening, or personally relevant were more likely to attract attention and interaction, even when participants expressed uncertainty about their accuracy. In several cases, participants described engaging with questionable content because it was entertaining or socially relevant rather than because they believed it was accurate. Participant F described this emotional response: "Definitely the shark. I'm afraid of sharks … If it's something about sharks, I just have to look, even though I don't really want to." This illustrates how fear could attract attention despite the participant's reluctance to engage with the content.

Familiarity also influenced credibility judgments. Participants often described trusting posts shared by familiar pages, known sources, or people within their social network. In the simulation, participants frequently clicked to view comments and reactions, suggesting that social feedback functioned as an important cue for interpreting posts. These patterns indicate that native platform cues such as comments, reactions, and perceived familiarity were more salient to many participants than direct content verification.

Passive and Low-Visibility Engagement Limited Nudge Exposure

Visible forms of engagement, particularly commenting and sharing, were relatively rare. Participants often described avoiding public interaction with news content because they did not want to appear controversial, provoke conflict, or be judged by others. Instead, they preferred lower-visibility behaviors such as liking, reading comments, scrolling, or observing the discussion without contributing. Participant C explained, "I rarely share. I'd almost say I never share things from Facebook. Not news. I don't comment either. But I do read the comments. I find that interesting." This illustrates how engagement often remained passive or occurred through low-visibility actions.

This pattern directly affected exposure to the accuracy prompt. Because the prompt appeared only when a participant attempted to share a post, many participants did not encounter it during the simulation. The low frequency of sharing therefore limited the behavioral reach of the prompt. This finding suggests that share-triggered nudges may miss users whose engagement with misinformation occurs mainly through passive or low-effort actions that still contribute to visibility and algorithmic circulation.

User Interpretation of the Two Nudges

The two nudges differed in visibility and interpretation. The social-norm message was more consistently noticed because it remained visible in the feed. Participant G reflected, "I personally thought it was well worded, and it made me pause and think a bit more when I started scrolling." This provides a direct account of how the persistent social-norm message was experienced as a subtle prompt for reconsideration. Participants generally described it as nonintrusive, although some interpreted it as being aimed at other users rather than themselves. This interpretation limited its personal relevance, even when participants agreed with the message in principle.

The accuracy prompt was encountered less frequently because it was triggered only by sharing attempts. Among participants who did encounter it, some reported that it made them pause or reconsider the action. The click-stream data also showed a small number of canceled shares after the prompt appeared. However, these observations should be interpreted cautiously. They indicate

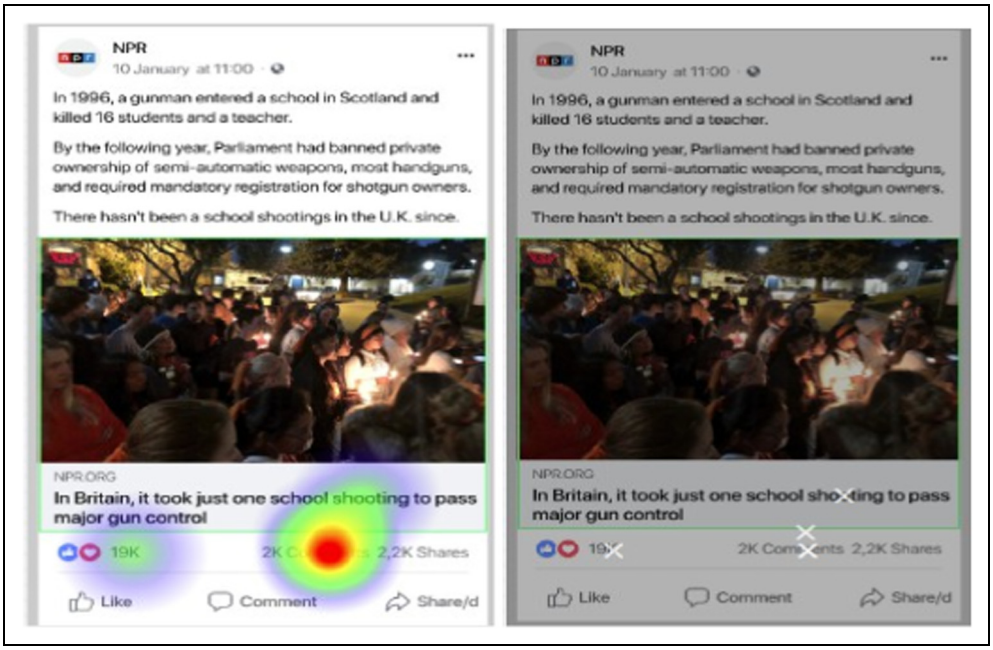


Figure 6. Participant's click patterns during the simulation.

moments of hesitation or reconsideration within the simulation, not evidence of improved accuracy discernment, belief change, or sustained reflective processing.

Behavioral interaction data further reflect differences between the control and intervention groups. Participants in the intervention group exhibited a wider range of interaction behaviors, including repeated clicks, canceled shares, and increased exploratory actions such as viewing articles or comment sections, as illustrated in Figure 6. These behaviors occurred alongside continued low levels of commenting and sharing overall. Taken together, the empirical findings indicate that the nudging elements coincided with brief interruptions in interaction patterns without being perceived as intrusive. Rather than producing consistent changes in engagement behavior, the nudges were associated with moments of hesitation and increased interaction variability within the simulated social media environment.

Overall, the nudges were generally perceived as acceptable and noncoercive. Their impact, however, was uneven. The social-norm message achieved higher visibility but was sometimes externalized as advice for others, while the accuracy prompt appeared more directly connected to reconsideration but reached fewer participants. These findings indicate that the artifact introduced limited moments of interactional friction, but its effectiveness was constrained by passive engagement patterns and the salience of native platform cues. The following section evaluates these empirical patterns against the design goals of the artifact rather than restating them as additional findings.

Artifact Evaluation

This section evaluates the artifact as the first design iteration within the DSR process (Table 4). The evaluation does not assess whether the nudges produced statistically significant behavioral change.

Table 4. Artifact Evaluation.

Evaluation Criterion	Evidence from the Study	Evaluation
Visibility	The social-norm message was more consistently noticed because it remained visible in the feed. The accuracy prompt was encountered less often because it appeared only during sharing attempts	One nudge was visible but sometimes externalized as advice for others; the other was more action-specific but had limited exposure
Usability and acceptability	Participants generally described both nudges as subtle, nonintrusive, and acceptable. They did not report that the nudges disrupted the simulated experience	The artifact preserved user autonomy and avoided perceptions of coercion
Behavioral reach	Sharing was rare, while passive behaviors such as liking, scrolling, and viewing comments were more common	A share-triggered prompt misses many users whose engagement occurs through low-visibility actions
Reflective potential	Some participants reported hesitation or reconsideration, and a small number of canceled shares occurred after the accuracy prompt	The evidence supports brief moments of possible reconsideration, not improved accuracy discernment or belief change
Design fit	Participants frequently used comments, reactions, and familiar sources as credibility cues. These native platform cues appeared more salient than the added social-norm message	Future nudges should be integrated more closely with the interaction points where users already seek social validation

Instead, it assesses whether the artifact met five criteria relevant to the study aim: visibility, usability and acceptability, behavioral reach, reflective potential, and fit with existing platform interaction patterns.

Overall, the artifact performed better as a low-disruption awareness intervention than as a reliable mechanism for changing misinformation engagement. Its strongest outcome was acceptability: participants generally perceived the nudges as transparent and noncoercive. Its weakest outcome was behavioral reach, especially for the accuracy prompt, which depended on a sharing action that many participants did not perform. The evaluation therefore supports the artifact as a useful first iteration but not as a validated solution. Future iterations should test nudges that appear during more common actions, such as liking, prolonged viewing, opening comments, or attempting to react to a post.

Discussion and Further Research

The empirical patterns observed in this study are consistent with prior research showing that misinformation engagement is shaped by emotional salience, cognitive shortcuts, overconfidence, familiarity, social validation, and expectations of platform responsibility. The study does not treat these mechanisms as new findings. Instead, its contribution lies in showing how these established mechanisms affect the reception and limits of interface nudges in practice. The findings suggest that overconfidence can lead users to view nudges as relevant mainly to others, that native platform cues such as comments and reactions may attract more attention than added normative messages, and that share-triggered prompts may have limited reach when users mostly engage through passive

or low-visibility actions. In this way, the study qualifies optimistic accounts of accuracy prompts and social-norm nudges by identifying interactional conditions under which such interventions may be accepted but only weakly effective. Building on the artifact evaluation, the discussion focuses on what these findings add to existing theory and design research, rather than repeating the descriptive results.

Interpreting the Findings Through Theory

To highlight theoretical insights and avoid redundancy, related findings are grouped into four thematic areas: cognitive and emotional drivers, social influence, trust and platform reliance, and user contradictions. Together, these interpretations clarify how the dual-system framework, social norms, and nudging theory interact in shaping misinformation engagement.

Fast Thinking, Biases of Cognition, and Emotional Factors. Participants appeared to rely heavily on fast and intuitive processing consistent with System 1 thinking. They often evaluated content based on surface-level cues such as layout, imagery, or familiarity, with limited reflective evaluation. This is consistent with previous findings that heuristic shortcuts and emotional values tend to prevail over thoughtful digital decision-making (Moravec et al., 2019, 2020; Pennycook et al., 2020b). Although it was anticipated to be the confirmation bias, the evidence also indicated the presence of overconfidence bias, anchoring, and framing effects. The participants tended to say that they were able to identify fake news via intuition or previous experience, at the same time they were describing their interaction with dubious content on the basis of superficial features. The illusion of personal invulnerability and the desire to shift the blame of misinformation issues on to others further supported cognitive overconfidence and insensitivity to perceived risk.

Emotions were also key factors. Even when they were aware that what they saw on the content might have been false, they still admitted to sharing it as it was either frightening, funny, or visually appealing. This is consistent with the findings of Dabbous and Barakat (2023) which revealed that emotionally appealing content will be shared at a higher rate regardless of the accuracy. It is also consistent with Pennycook et al''s (2021) argument that engagement with misinformation is often associated with inattention rather than deliberate evaluation.

This finding does not challenge prior work showing that emotional and intuitive processing contribute to misinformation engagement. Rather, it qualifies the practical use of nudging interventions by showing that emotional salience can remain dominant even when users notice or approve of accuracy-oriented cues. The implication is that a nudge may be acceptable and theoretically well-aligned with dual-process accounts yet still fail to redirect engagement when the content is entertaining, personally relevant, or affectively intense at the point of interaction.

Social Cues, Peer Influence, and Norm Conflicts. Participants frequently used social cues such as comments and reactions to judge credibility, reflecting the influence of descriptive norms. Peer behavior appeared more persuasive than injunctive norms promoting responsible sharing. This aligns with prior work showing that peer behavior strongly shapes online decision-making (Andı & Akesson, 2021; Prike et al., 2024). Although participants claimed to value accuracy, their actions more often followed peer cues than norm-based messages about responsible behavior. These dynamics reveal that the visibility and placement of norm-based interventions are crucial. Messages that appear outside users' main attention flow, such as at the top of a feed, risk being overlooked or dismissed as irrelevant. Integrating social cues where attention is naturally focused, for instance in comment sections, could increase effectiveness but raises ethical and practical

challenges related to transparency and scalability. Effective design must therefore align normative messaging with the spaces where users already seek validation.

The findings also extend prior work on social-norm interventions by distinguishing between designed normative cues and the platform's native social signals. Participants were often more attentive to comments, reactions, and perceived source familiarity than to the added social-norm message. This suggests that social-norm nudges do not operate in isolation; they compete with existing interface signals that already communicate popularity, familiarity, and social endorsement. Future designs may therefore need to embed normative cues closer to the spaces where users already seek social validation, while still preserving transparency and user autonomy.

Trust, Familiarity, and Platform Delegation. Participants commonly outsourced trust to platforms, assuming algorithms would filter misinformation and prioritize credible sources. This misplaced trust mirrors findings by French et al. (2025) and Lazer et al. (2018), who note that users tend to overestimate the reliability and gatekeeping function of their digital environments. While participants expressed a desire for platform-led solutions, few recognized existing fact-checking tools, echoing prior research showing that credibility cues often go unnoticed (Moravec et al., 2019). Familiarity bias further reinforced this trust. Participants relied on known pages or friends as indicators of reliability, illustrating how repeated exposure and social proximity can enhance perceived truthfulness (Pennycook et al., 2020a). In the simulation, this bias may have been underestimated due to the lack of personal network connections, suggesting even stronger effects in real-world conditions. Overall, these findings underscore the importance of designing cues that are both visible and contextually relevant within users' existing trust models.

Overlaps, Tensions, and User Contradictions

Participants' reflections revealed contradictions between stated values and actual behaviors. While many distanced themselves from misinformation, several acknowledged acting against their intentions when faced with emotionally or visually engaging content. These moments of introspection suggest that interventions, even subtle ones, can trigger self-awareness and cognitive friction. However, this awareness did not always translate into observable differences in interaction patterns, highlighting the gap between recognition and action. Participants often rationalized sharing false or exaggerated content as harmless or entertaining. These observations are consistent with prior findings that misinformation sharing may be motivated by perceived social or prosocial value rather than belief in accuracy (Balakrishnan et al., 2021). Some viewed nudges as helpful yet primarily for other users. This self-other bias, coupled with selective trust in peers and skepticism toward moderation, reveals the psychological complexity of misinformation behavior. Users simultaneously resist control while desiring protection, a paradox that design interventions must navigate carefully.

Design Implications

The study highlights the need for nuanced, ethically grounded interventions that foster reflective engagement without interrupting the user experience. Nudges should integrate seamlessly into natural interaction flows, balancing intuitive cues with moments of reflective processing (Moravec et al., 2020). Because most users engage passively through scrolling, liking, or observing, interventions should also address low-effort behaviors that indirectly reinforce algorithmic visibility (Butler et al., 2024). Design tone and framing are equally critical. Nudges perceived as moralizing or coercive may trigger resistance, while transparent and autonomy-supportive cues encourage acceptance.

Participant feedback indicates that both tested nudges were acceptable, but would benefit from improved clarity, timing, and contextual integration. Future iterations should aim for subtle but noticeable delivery, ensuring ethical transparency while maintaining engagement.

Refining the Social Norm Message. The social norm message could be more dynamic by responding to user behavior rather than remaining static. For example, it might appear when users pause on a post or begin scrolling, then collapse into an optional icon linking to credibility resources. Shifting the phrasing from moral guidance to shared vulnerability may also improve engagement. For example, a statement could present a verified, context-specific statistic such as "[X%] of users share misinformation unintentionally. Pause before interacting." This framing could then emphasize collective risk and encourage empathy rather than judgment.

Refining the Accuracy Prompt. The accuracy prompt showed potential but would benefit from greater visibility and optional depth, such as linking to a credibility checklist or offering context on why accuracy matters. Subtle wording adjustments, for example "Are you confident this post is accurate?", could activate reflection without adding friction. Testing these refinements in more realistic settings and among diverse user groups would provide valuable insight into long-term effects.

Future Research

Several limitations qualify these findings. The study rests on a single simulated session with a small, self-selected sample, no longitudinal follow up, and no iterative design testing; the proposed nudge refinements are therefore untested hypotheses rather than evidenced improvements. The Maze/Figma simulation lacked participants' real social networks and could not fully reproduce real-feed dynamics. Its constrained sharing environment therefore limits the ecological validity and generalizability of the findings. Ethically, nudging raises questions of manipulation, autonomy, and transparency: more salient cues may be more effective, but risk being perceived as coercive. The interventions tested were designed to be transparent and autonomy-preserving, and informed consent procedures were followed. Real-world deployment would additionally depend on platform cooperation and governance, which lies beyond the scope of this study.

The findings emphasize the value of studying misinformation engagement in ecologically valid environments. Future research should expand sample diversity and explore longitudinal effects of repeated exposure to nudges. Additional work is needed to examine interventions that target passive engagement, emotional triggers, and network-based dynamics. By combining behavioral insights with ethical design principles, future studies can advance scalable, autonomy-preserving interventions that strengthen reflective engagement in digital environments.

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Ethical Considerations

This study adhered to institutional ethical guidelines and ensured compliance with GDPR in all aspects of data collection, handling, and storage. According to the applicable institutional policies and the nature of this research, approval from an Institutional Review Board (IRB) or Research Ethics Committee was not required.

Consent to Participate

Participation in the study was entirely voluntary. Prior to engaging in the interview, simulation, or survey, all participants were provided with clear information about the purpose of the study, how their data would be used, and their right to withdraw at any time without consequence. Informed consent was obtained from all interview participants through signed consent forms. Participation in the simulation and surveys was fully anonymous, and no personally identifiable data were collected or stored at any point.

Funding

The authors received no financial support for the research, authorship, and/or publication of this article.

Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Data Availability Statement

The data that support the findings of this study are available upon request.

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